

Photophobia

Photophobia is a medical symptom of abnormal intolerance to visual perception of light.^[1] As a medical symptom photophobia is not a morbid fear or phobia, but an experience of discomfort or pain to the eyes due to light exposure or by presence of actual physical sensitivity of the eyes,^[2] though the term is sometimes additionally applied to abnormal or irrational fear of light such as heliophobia.^[3] The term *photophobia* comes from the Greek φῶς (*phōs*), meaning "light", and φόβος (*phóbos*), meaning "fear".

Photophobia	
Specialty	<u>Ophthalmology</u>

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Causes

Patients may develop photophobia as a result of several different medical conditions, related to the eye, the nervous system, genetic, or other causes.

Photophobia may manifest itself in an increased response to light starting at any step in the visual system, such as:

- Too much light entering the eye. Too much light can enter the eye if it is damaged, such as with corneal abrasion and retinal damage, or if its pupil(s) is unable to normally constrict (seen with damage to the oculomotor nerve).
- Due to albinism, the lack of pigment in the colored part of the eyes (irises) makes them somewhat translucent. This means that the irises can't completely block light from entering the eye.
- Overstimulation of the photoreceptors in the retina
- Excessive electric impulses to the optic nerve
- Excessive response in the central nervous system

Common causes of photophobia include migraine headaches, TMJ, cataracts, Sjögren syndrome, mild traumatic brain injury (MTBI), or severe ophthalmologic diseases such as uveitis or corneal abrasion.^[4] A more extensive list follows:

Eye-related

Causes of photophobia relating directly to the eye itself include:

- Achromatopsia^[5]
- Aniridia^[6]
- Anticholinergic drugs may cause photophobia by paralyzing the iris sphincter muscle.
- Aphakia^[7]
- Blepharitis^[8]
- Buphthalmos^[6]
- Cataracts^[6]
- Coloboma
- Cone dystrophy^[6]
- Congenital abnormalities of the eye^[6]
- Viral conjunctivitis^[9]
- Corneal abrasion^[6]
- Corneal dystrophy^[6]
- Corneal ulcer^[10]
- Disruption of the corneal epithelium, such as that caused by a corneal foreign body or keratitis^[6]
- Ectopia lentis^[6]
- Endophthalmitis^[6]
- Eye trauma caused by disease, injury, or infection such as chalazion, episcleritis, keratoconus, or optic nerve hypoplasia^[6]
- Hydrophthalmos, or congenital glaucoma^[6]
- Iritis^[6]
- Isotretinoin has been associated with photophobia^[11]
- Optic neuritis^[6]
- Pigment dispersion syndrome
- Pupillary dilation (naturally or chemically induced)^[7]
- Retinal detachment
- Scarring of the cornea or sclera^[6]
- Uveitis^[6]

Nervous-system-related

Neurological causes for photophobia include:

- Autism spectrum disorders^[12]
- Chiari malformation
- Dyslexia^[13]
- Encephalitis^[6] including myalgic encephalomyelitis^[14]
- Meningitis^[6]
- Trigeminal disturbance causes central sensitization (hence, multiple other associated hypersensitivities). Causes can be bad bite, infected tooth, etc.^[15]
- Progressive supranuclear palsy, where photophobia can sometimes precede the clinical diagnosis by years^[16]
- Subarachnoid haemorrhage^[17]
- Tumor of the posterior cranial fossa^[6]
- Visual snow along with many symptoms.^{[18][19]}

Other causes

- Ankylosing spondylitis^[20]

- [Albinism](#)^[21]
- [Ariboflavinosis](#)^[22]
- [Benzodiazepines](#)^{[23][24]}
- [Chemotherapy](#)^[6]
- [Chikungunya](#)^[25]
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- [Migraine](#)^[30]
- [Rabies](#)^[31]
- [Tyrosinemia type II](#)^[6]
- [Superior canal dehiscence syndrome](#)^[32]

Treatment

Treatment for light sensitivity addresses the underlying cause, whether it be an eye, nervous system or other cause. If the triggering factor or underlying cause can be identified and treated, photophobia may disappear. Tinted glasses are sometimes used.^[33]

Artificial light

People with photophobia may feel eye pain from even moderate levels of artificial light and avert their eyes from artificial light sources. Ambient levels of artificial light may also be intolerable to persons afflicted with photophobia such that they dim or remove the light source, or go into a dimmer lit room, such a one lit by refraction of light from outside the room. Alternatively, they may wear dark [sunglasses](#), sunglasses designed to filter [peripheral](#) light, and/or wide-brimmed [sun hats](#) or [baseball caps](#). Some types of photophobia may be helped with the use of precision tinted lenses which block the green-to-blue end of the light spectrum without blurring or impeding vision.^{[34][35]}

Other strategies for relieving photophobia include the use of tinted contact lenses and/or the use of prescription eye drops that constrict the pupil, thus reducing the amount of light entering the eye. Such strategies may be limited by the amount of light needed for proper vision under given conditions, however. Dilating drops may also help relieve eye pain from muscle spasms or seizures triggered by lighting/migraine, allowing a person to "ride out the migraine" in a dark or dim room. A paper by Stringham and Hammond, published in the *Journal of Food Science*, reviews studies of effects of consuming [Lutein](#) and [Zeaxanthin](#) on visual performance, and notes a decrease in sensitivity to glare.^[36]

Disability

Photophobia may preclude or limit a person from working in places where lighting is used, unless the person is able to obtain a reasonable accommodation like being allowed to wear tinted glasses when the employee [dress code](#) prohibits such glasses otherwise (which may be required to be provided by an employer under the [Americans with Disabilities Act](#)). Some people with photophobia may thereby be better able to work at night or be more easily accommodated in the workplace at night.

Outdoor night lighting may be equally offensive for persons with photophobia, however, given the wide variety of bright lighting used for illuminating residential, commercial and industrial areas, such as [LED](#) (light-emitting diode) lamps.^{[37][38]}

The increasing popularity of "overpoweringly intense" LED headlights being used on "[pickups](#) and [S.U.V.s](#)" has prompted more frequent reports of photophobia among motorists, cyclists, and pedestrians.^[39]

See also

- Photic sneeze reflex, a medical condition by which people exposed to bright light sneeze
- Photosensitivity in humans

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